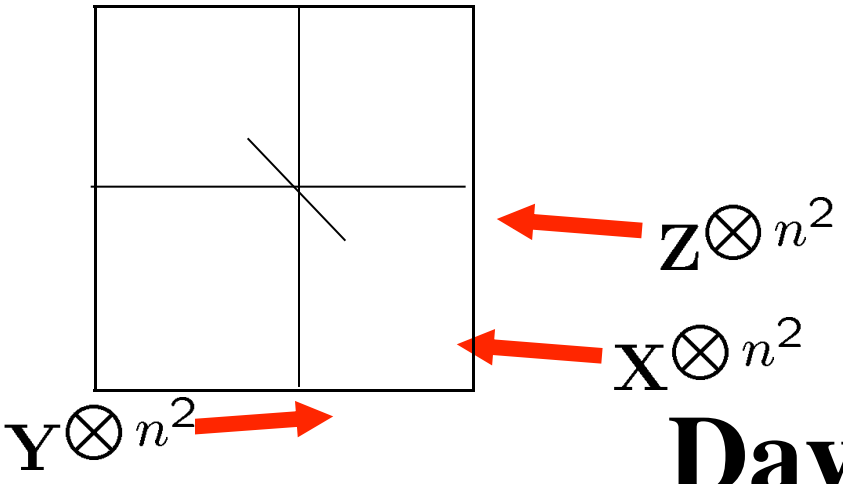
There's Plenty of Room at the Top

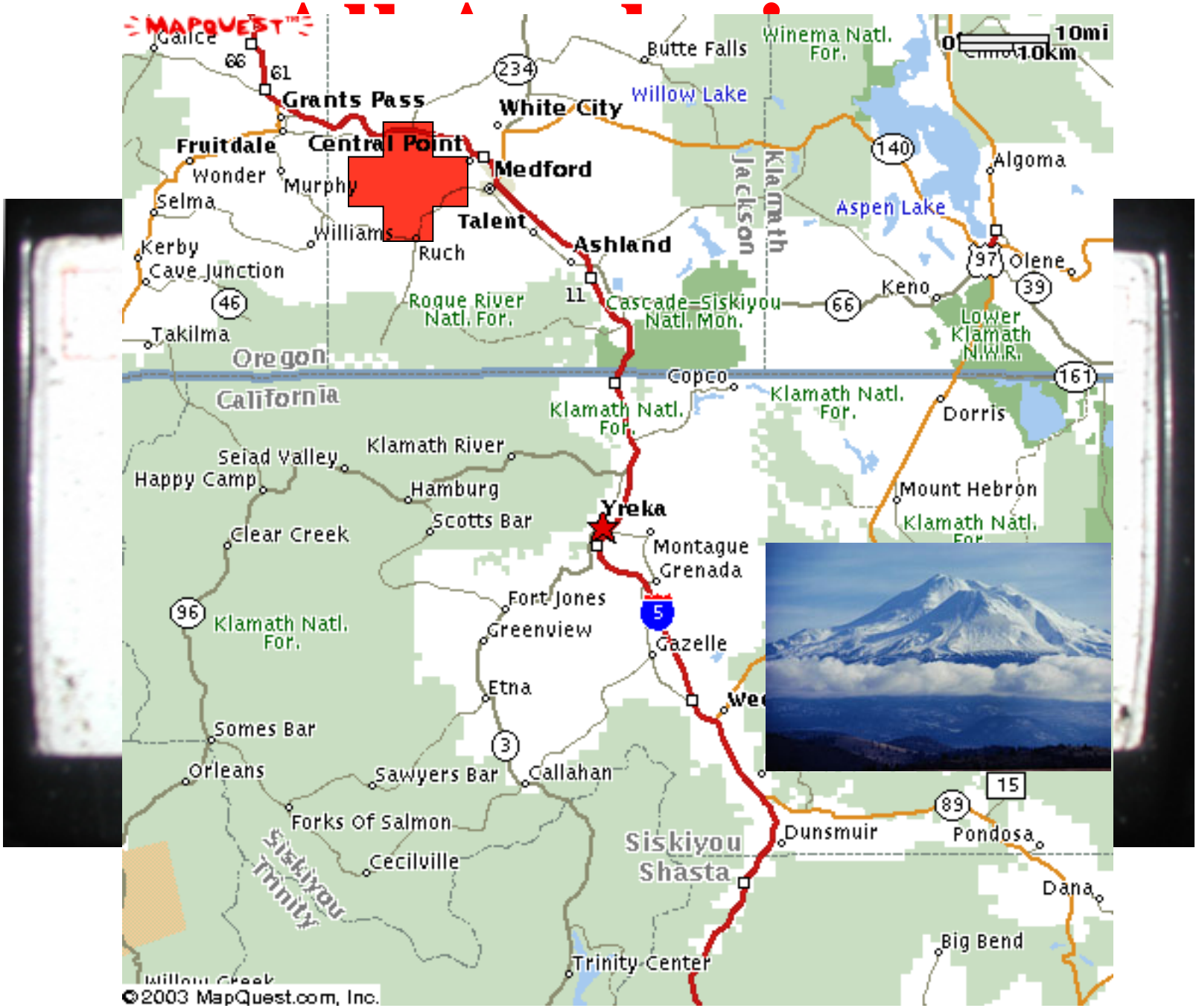
An Invitation to Help Start the Quantum Computing Revolution



Dave Bacon

Institute for Quantum Information

Caltech


MAPQUEST
Gaice
66
61
Grants Pass
234
Butte Falls
Winema Natl. For.
10mi
10km
White City
Willow Lake
Fruitdale
Central Point
Medford
Wonder
Murphy
Selma
Talent
Ashland
Williams
Ruch
Kerby
Cave Junction
46
Takilma
Roque River Natl. For.
11
Cascade-Siskiyou Natl. Mon.
Jackson
Klamath
140
Algoma
Aspen Lake
97
Olene
Keno
66
39
Lower Klamath N.W.R.
Oregon
California
Copco
Klamath Natl. For.
161
Dorris
Seiad Valley
Klamath River
Happy Camp
Hamburg
Scotts Bar
Yreka
Mount Hebron
Clear Creek
Montague
Grenada
Fort Jones
96
Klamath Natl. For.
Greenview
5
Gazelle
Etna
Somes Bar
3
Orleans
Sawyers Bar
Callahan
15
89
Forks Of Salmon
Cecilville
Siskiyou Trinity
Siskiyou Shasta
Dunsmuir
Pondosa
Dana
Big Bend
Trinity Center
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The Big Two

(what everyone should be working on in quantum computing)



Quantum Algorithms

What Will be THE Quantum Computing Technology?

Rant mode ON

Physics and Computation

<u>The Computer Science Assumption</u>: *robust* digital storage and digital manipulation of information is possible.

In this NOISY (ANALOG) world HOW can anything be DIGITAL?

Theory: Von Neumann showed methods for fault-tolerant computation

Practice: no need for these methods because

THERE EXIST SYSTEMS WHOSE PHYSICS GUARANTEES ROBUST CLASSICAL COMPUTATION

<u>THE BILLION DOLLAR QUANTUM QUESTION</u>

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?

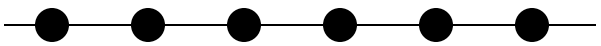
The Physics Guarantee

Rant mode SHUTTING OFF

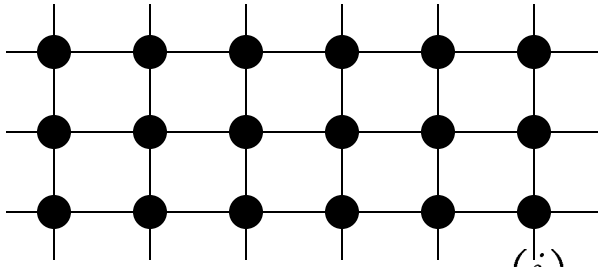
Classical Ising memory

$$\mathbf{H} = -J \sum_{\langle i,j \rangle} \boldsymbol{\sigma}_z^{(i)} \boldsymbol{\sigma}_z^{(j)}$$

One dimensional Ising model



Two dimensional Ising model

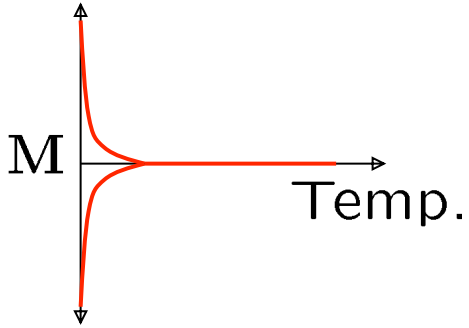


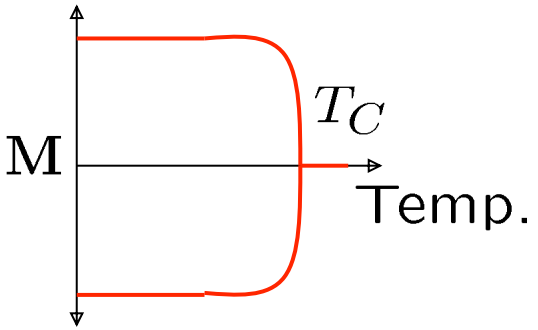
Redundancy code: Stabilizer code with elements $\boldsymbol{\sigma}_z^{(i)} \boldsymbol{\sigma}_z^{(j)}$

Codewords are ground states $|0 \cdots 0\rangle$, $|1 \cdots 1\rangle$

Order parameter: total magnetization, $\mathbf{M} = \sum_i \boldsymbol{\sigma}_z^{(i)}$

indication that the information is "still there"

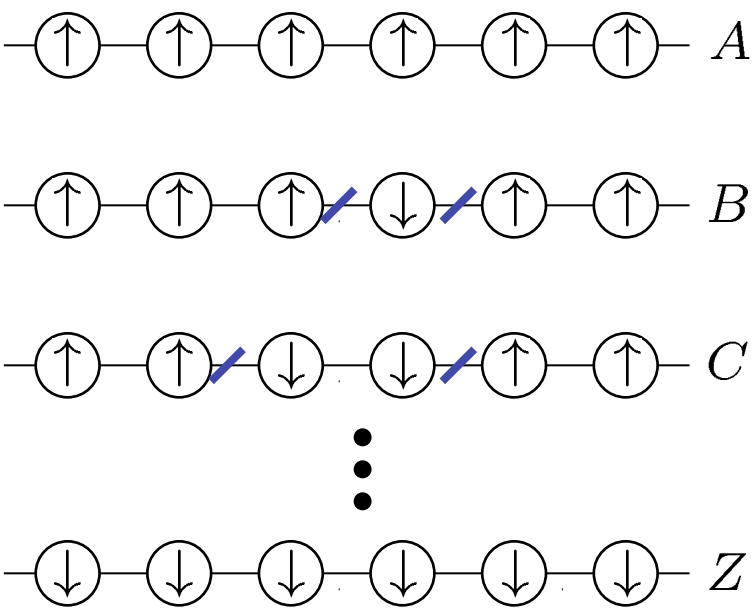


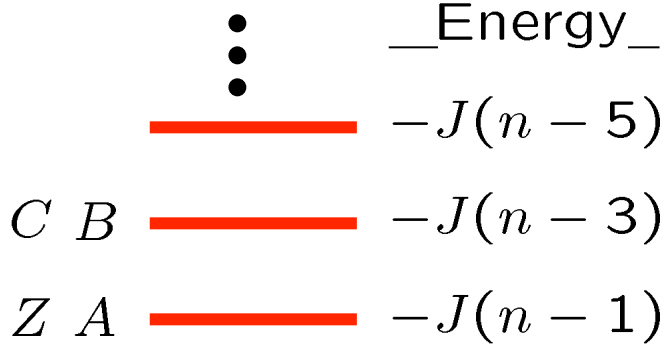


Don't build memory in 1D

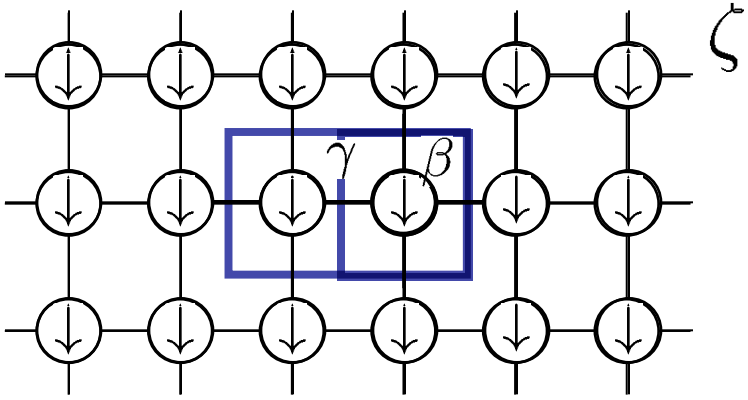
2D memories much better

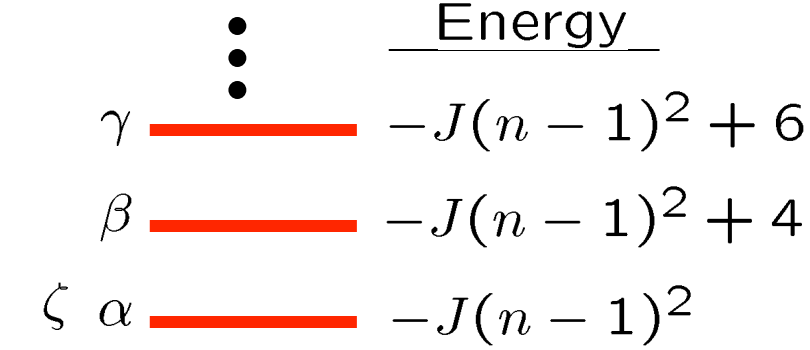
Self Fixing





Excitations pointlike.
Energy cost constant.
disordering entropy $\gg$ energy

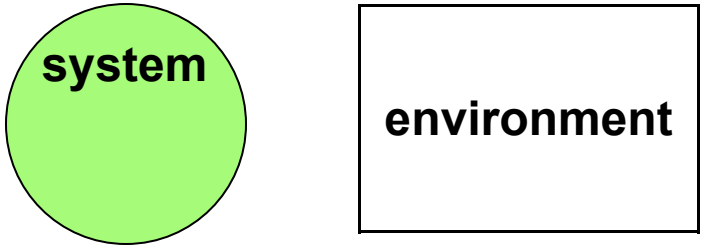




Excitations stringlike.
Energy cost grows like perimeter.
energy $\gg$ disordering entropy

The Quantum Problem

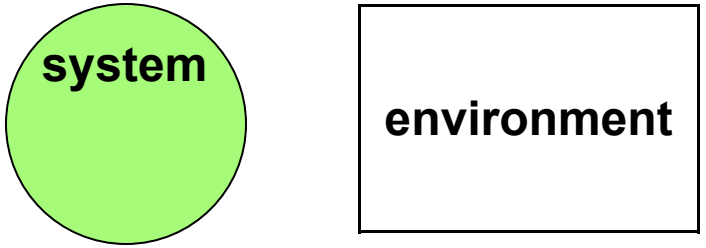
Classical



Vini $\otimes$ Wini initial

$\sum_a P_a$ Va $\otimes$ Wa final

Quantum



$|\psi_{sys}\rangle \otimes |\psi_{env}\rangle$

$|\psi\rangle = \sum_a c_a |a_{sys}\rangle \otimes |a_{env}\rangle$

Other Issues of Control / Noise SIMILAR.

Current State: like Von Neumann
theory of fault-tolerant quantum computation

Two Paths

Hard Drive



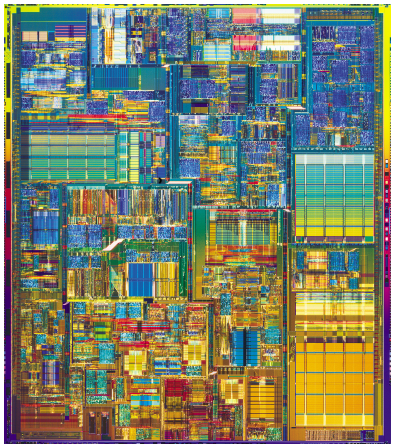
Coding: majority vote of magnetism

Error correction: local energy minimization

Fault-tolerant: macro applied field.

More than storage?

Transistor



Coding: majority vote of current

Error correction: amplification

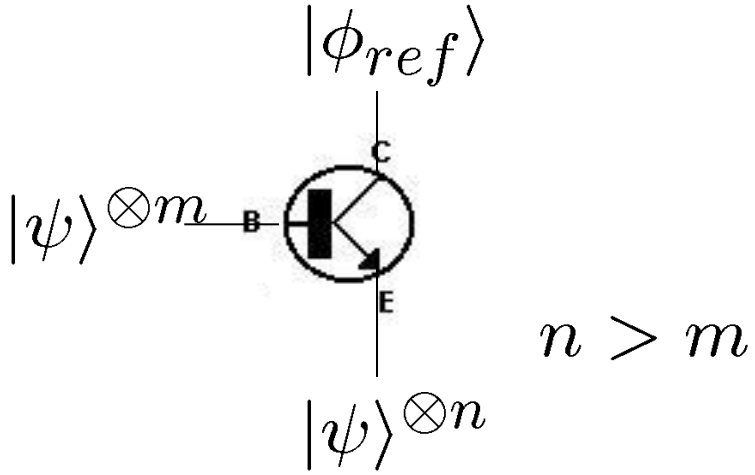
Fault-tolerant: conducting / insulating phase transition

No Quantum Transistor



Quantum Cloning Machine

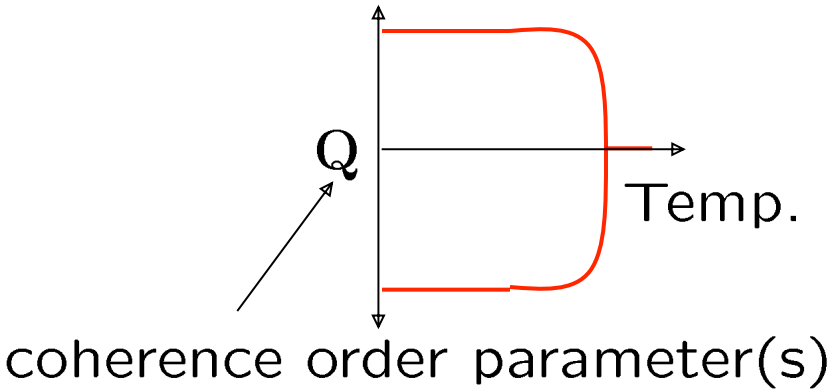
"A single quantum cannot be cloned," Wootters and Zurek, Nature, 1982



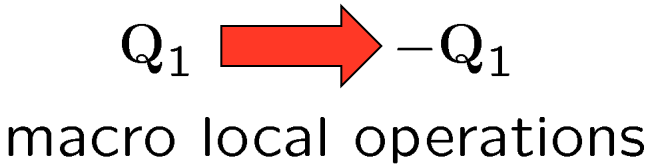
The Quantum Hard Drive?

Alexei Kitaev, Michael Freedman, John Preskill, …

Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



1. Coherence preserving.



2. Accessible Fault-Tolerant Operations

$$\left\langle G_1, \ldots, G(n^k) \right\rangle \cong U(c^n)$$

3. Universality

Towards Quantum Hard Drives

Kitaev – “computation by anyons”
- first suggestion of natural fault-tolerance (1997).
- fault-tolerant universal quantum computation when actively used
- gapped, but thermal fluctuations disorder system
- >2 body interactions, many-many level systems

Dennis, Kitaev, Landahl, and Preskill – “four dimensional code”
- local four dimensional code
- should have natural ordered phase
- >2 body interactions

Freedman – “quantum loop gas”
- 2-dimensional
- analogy with classical loop gas argues for ordered phase
- >2 body interactions

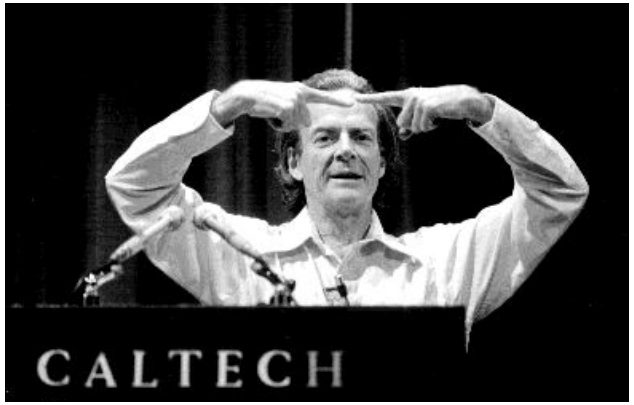
Interactions

What can be done with two-body interactions?



interactions from proposed quantum computers

There’s Plenty of Room at the Bottom



Richard P. Feynman

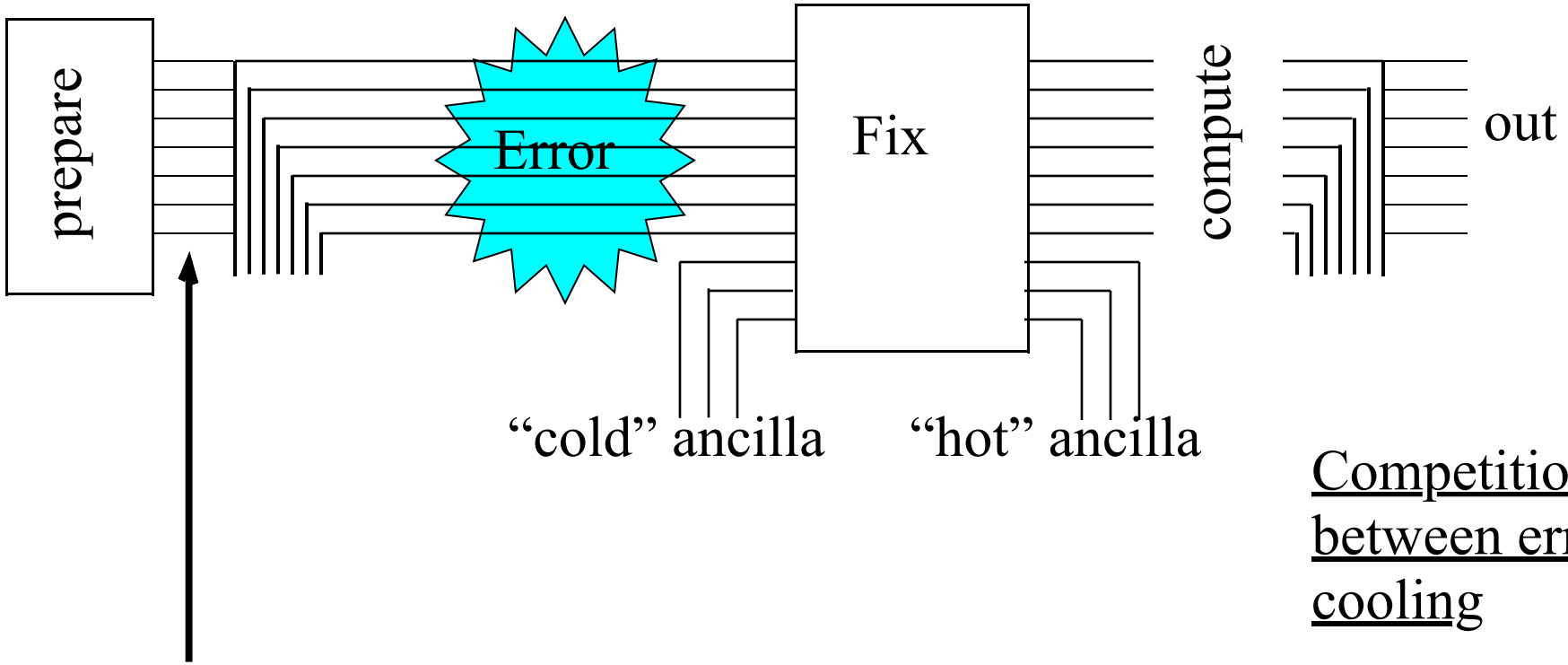
“There's plenty of room at the bottom: an invitation to open up a new field of physics,” talk given approximately 1960

The Computer Revolution

Macro ⇄ **Meso** ⇄ **Micro**

The Quantum Computer Revolution? NO! NO! NO!

Old School



Competition between errors a cooling

Quantum error correcting code

$$\mathcal{H} \subset \mathcal{V}^{\otimes n}$$

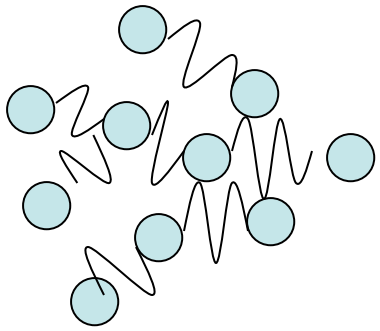
$$|i_L\rangle = \sum_\alpha c_{\alpha,i}|\alpha\rangle, \alpha \in \mathcal{V}^{\otimes n}$$

Error correcting criteria:

$$\langle i_L|\mathbf{E}_b^\dagger \mathbf{E}_a|j_L\rangle = c_{ab}\delta_{i,j}$$

If criteria holds, then fixing procedure possible

Into The Fray



Thermodynamic limit

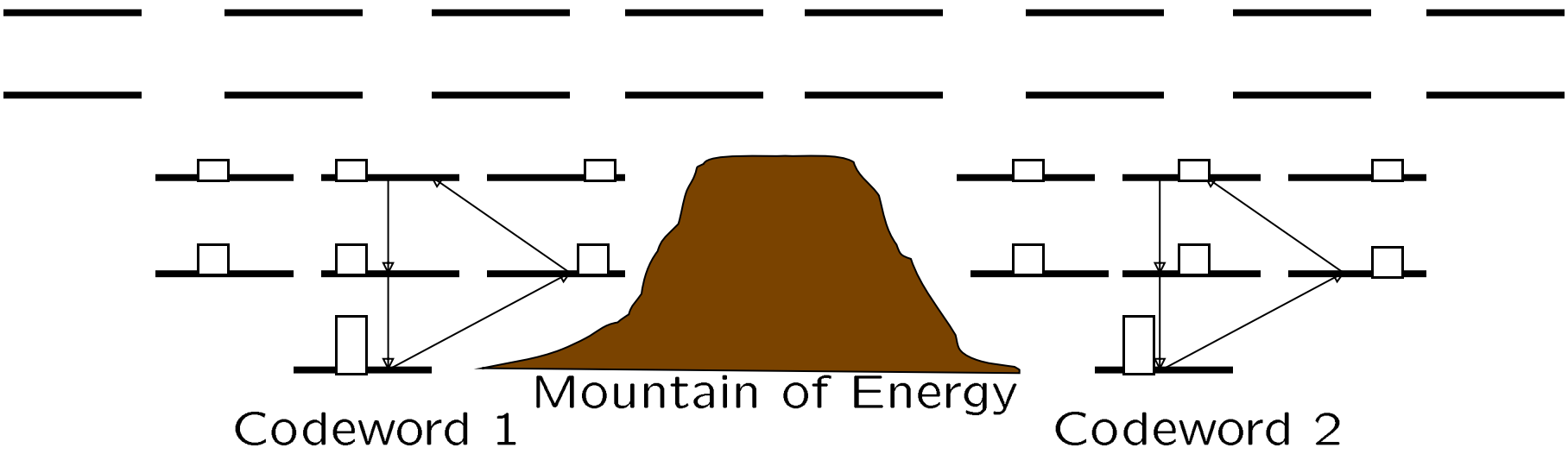
$$N \to \infty$$

guarantees robustness

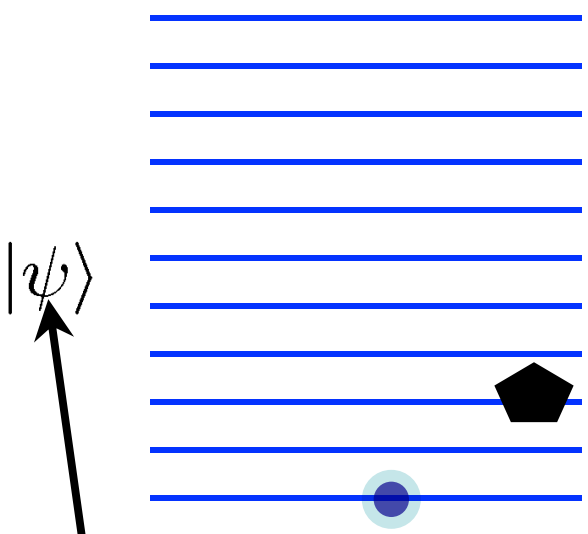
Strongly Interacting Many Body Quantum System

Hamiltonian $\mathbf{H}$

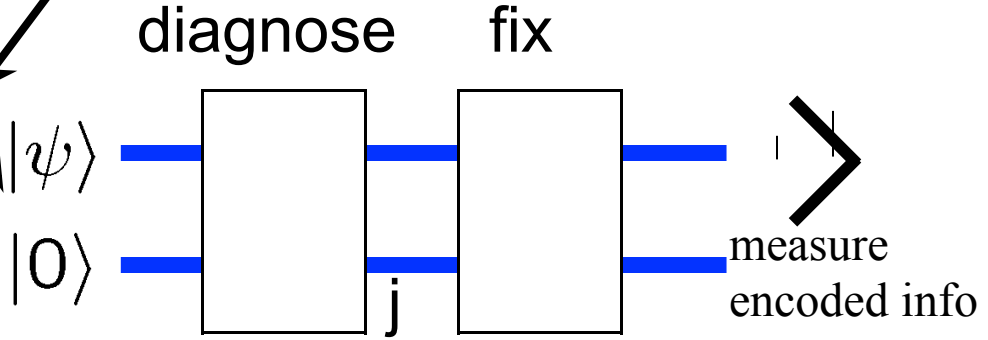
energy eigenstates



Quantum Error Correcting Order Parameter



The quantum information is still here. How do we see it?



Encoded quantum information

$$|\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$$

$$\mathbf{Q}_s = \sum_j \mathbf{L}_s \mathbf{F}_j \mathbf{D}_j$$

$\mathbf{L}_s$ logical encoded operation

$\mathbf{D}_j$ projector onto jth errored subspace

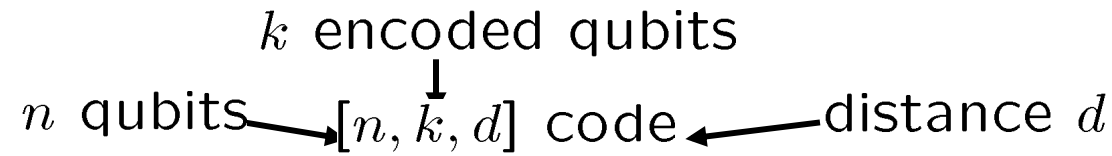
$\mathbf{F}_j$ fixing operator for jth errored subspace

A Simple Start

"To be an Error and to be Cast out is Part of [Nature's] Design."

-*William Blake*

$$\mathbf{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \mathbf{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \mathbf{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



[4,2,2] quantum error detecting code:

$$\mathbf{S}_1 = \mathbf{XXXX} \quad \mathbf{S}_2 = \mathbf{ZZZZ} \qquad \mathbf{S}_i|\psi\rangle = |\psi\rangle$$

Encoded operations on encoded ($\mathbf{S}_i|\psi\rangle = |\psi\rangle$) two qubits

$$\mathbf{X}_1 = \mathbf{XXII},\ \mathbf{Z}_1 = \mathbf{IZZI}$$

$$\mathbf{X}_2 = \mathbf{IXXI},\ \mathbf{Z}_2 = \mathbf{ZZII}$$

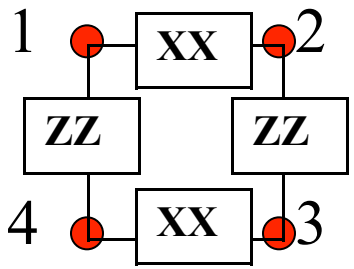
Complete basis: $|S_1, S_2, Z_1, Z_2\rangle,\ S_i, Z_i \in \{\pm 1\}$

$\mathbf{PS}_i = -\mathbf{S}_i\mathbf{P}$ for some i

$\langle i_L|\mathbf{P}|j_L\rangle = 0$, $\mathbf{P}$ single qubit operator

Supercoherence

4 qubit system



$$S_1 = XXXX$$
$$S_2 = ZZZZ$$
$$X_1 = XXII,\ Z_1 = IZZI$$
$$X_2 = IXXI,\ Z_2 = ZZII$$

System Hamiltonian: $H = XXII + IIXX + ZIIZ + IZZI$

Note symmetries: $[H, S_i] = [H, X_2] = [H, Z_2] = 0$

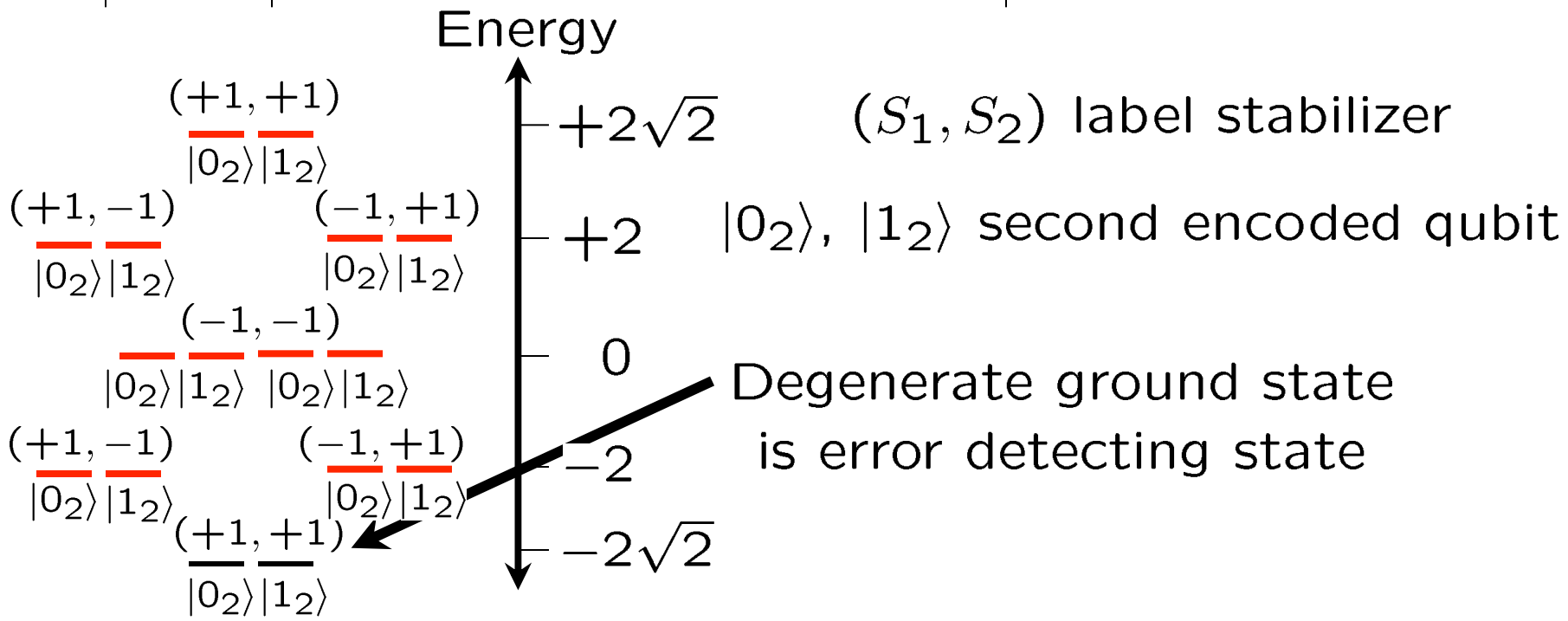
$$H = X_1(I + S_1) + Z_1(I + S_2)$$

Does not involve 2nd encoded qubit.

$\Downarrow$

every energy level at least two-fold degenerate

S_1	S_2	$\mathbf{H} = \mathbf{X}_1(1+S_1) + \mathbf{Z}_1(1+S_2)$	Eigenvalues
$+1$	$+1$	$2(\mathbf{X}_1 + \mathbf{Z}_1)$	$\pm 2\sqrt{2}$
$+1$	-1	$2\mathbf{X}_1$	± 2
-1	$+1$	$2\mathbf{Z}_1$	± 2
-1	-1	$\mathbf{0}$	0



Supercoherence

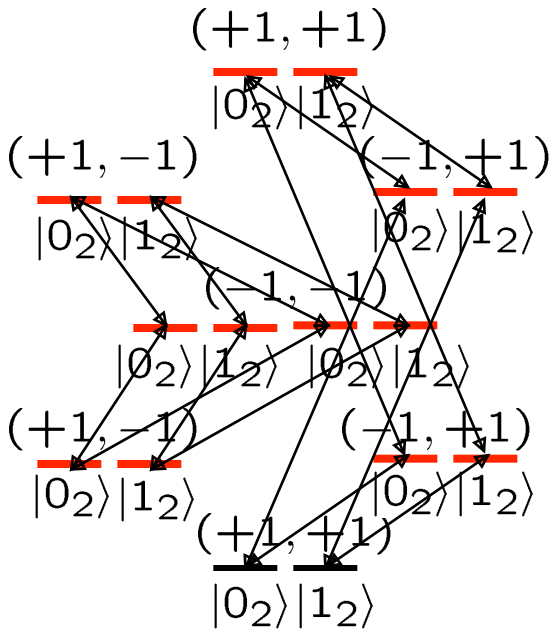
$$\mathbf{H} = \mathbf{X}_1(1 + S_1) + \mathbf{Z}_1(1 + S_2)$$

$\mathbf{S}_i\mathbf{P} = -\mathbf{P}\mathbf{S}_i$ for some i $\quad$ $\mathbf{P}$ single qubit operator

ALL single qubit errors take degenerate ground state to higher energy levels.

Single qubit errors change value of (S1,S2) and hence take ground state to higher energy level.

Effect of **ZIII**

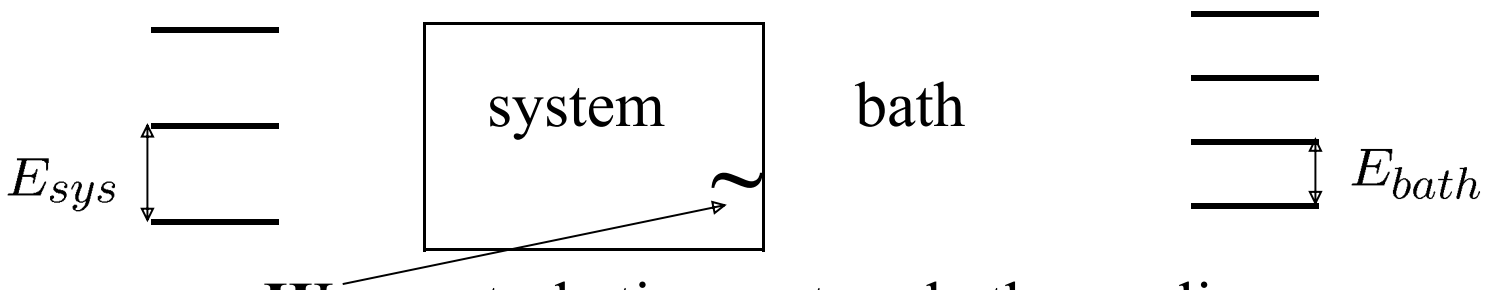


$$\underbrace{(+1,}_{|0_2\rangle}\ \underbrace{+1)}_{|1_2\rangle}$$

✗ **FORBIDDEN**

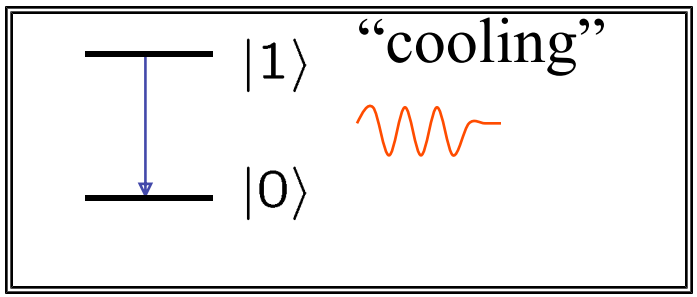
D. Bacon, Ph.D. thesis, U.C. Berkeley, 2001

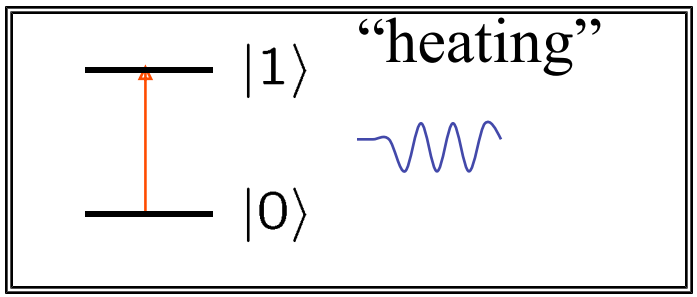
Making Decoherence Work Hard

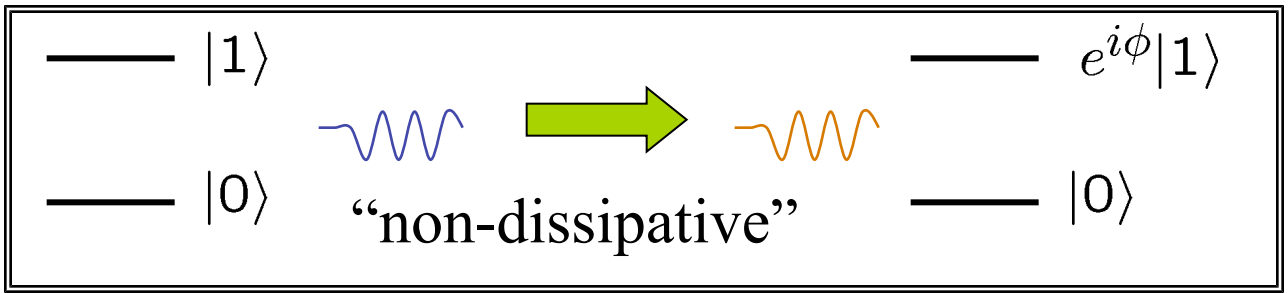


HI = perturbative system-bath coupling

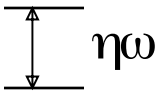
HI energy small compared to Esys and Ebath implies decoherence dominated by pathways that *conserve unperturbed energies* (essentially rotating wave approximation)







two-level atom radiatively coupled to a thermal reservoir:



$$\frac{\partial \rho}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\sigma_-\sigma_+\rho + \rho\sigma_-\sigma_+ - 2\sigma_+\rho\sigma_-\right]$$

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\sigma_+\sigma_-\rho + \rho\sigma_+\sigma_- - 2\sigma_-\rho\sigma_+\right]$$

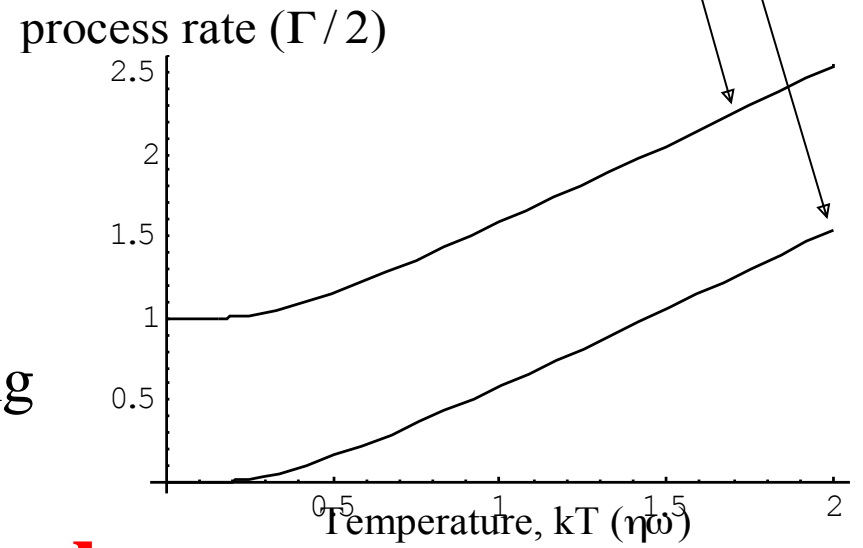
$$-\phi\left[\rho - \sigma_z\rho\sigma_z\right]$$

"heating"

"cooling"

"non-dissipative"

$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\eta\omega}{kT}\right) - 1}$$



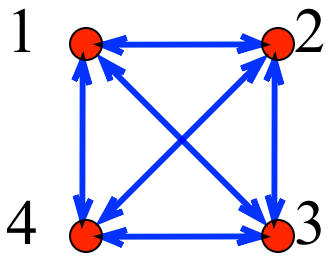
At low bath temperatures, heating disappears……..

Supercoherence made all quantum errors "heating"!

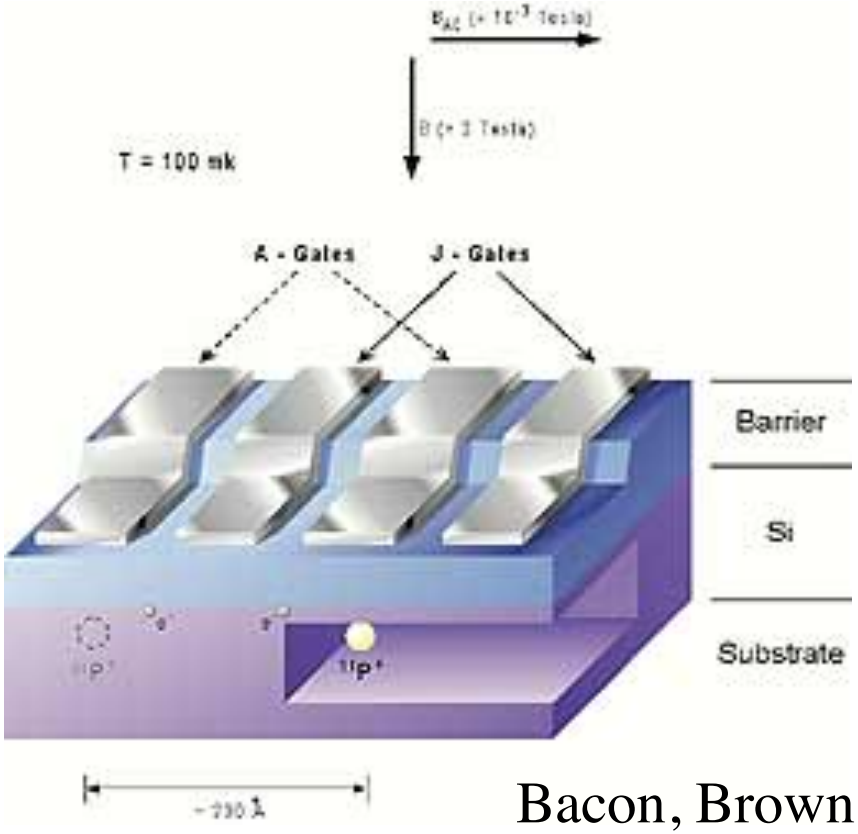
In a Few Years Time…

Four spin systems with equal strength exchange interactions between all spins:

$$\mathbf{H} = \frac{\Delta}{2}\sum_{i,j=1}^{4} \hat{s}_i \cdot \hat{s}_j = \frac{\Delta}{2}\hat{\mathbf{S}}^2 + c\mathbf{I}$$



Supercoherent qubit



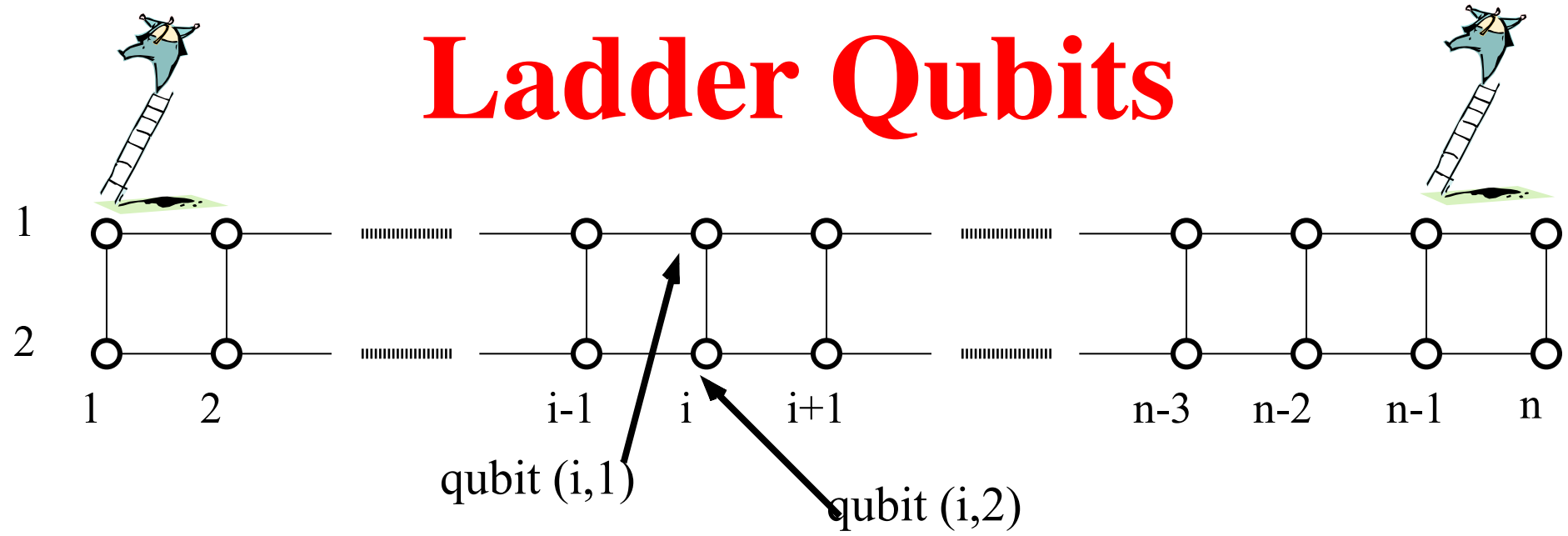
single P atoms in Silicon

exchange controlled by voltage mediated barriers

Bacon, Brown, and Whaley, PRL **87**, 247902 (2001)

DiVincenzo, Bacon, Kempe, Burkard, and Whaley, Nature **408,** 399 (2001)

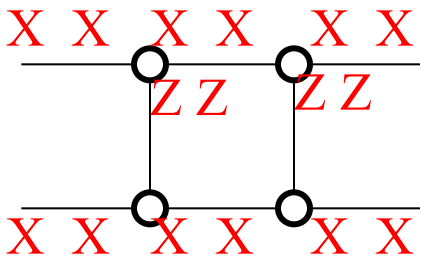
Ladder Qubits



$\sigma_\alpha^{(i,j)} \equiv \alpha^{\text{th}}$ Pauli operator acting on the (i, j) qubit tensored with I on all other qubits

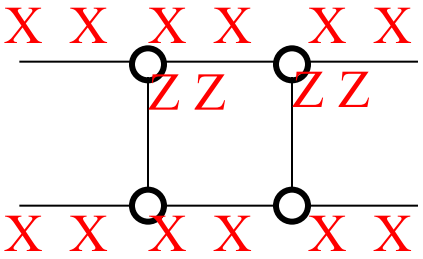
The Hamiltonian

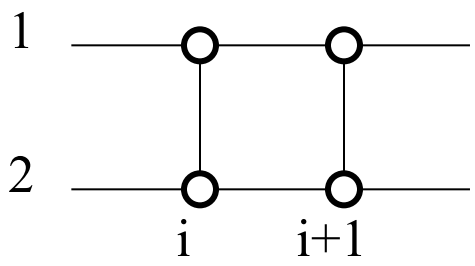
$$H = -\omega \left[\sum_{i=1}^{n} \sigma_Z^{(i,1)} \sigma_Z^{(i,2)} + \sum_{i=1}^{n-1} \left(\sigma_X^{(i,1)} \sigma_X^{(i+1,1)} + \sigma_X^{(i,2)} \sigma_X^{(i+1,2)} \right) \right]$$



D. Bacon, Ph.D. thesis, U.C. Berkeley, 2001

Basis Change





New basis:

$$\mathbf{X}_1^{(i)} = \sigma_x^{(i,1)}, \mathbf{Z}_1^{(i)} = \sigma_z^{(i,1)}\sigma_z^{(i,2)}$$

$$\mathbf{X}_2^{(i)} = \sigma_x^{(i,1)}\sigma_x^{(i,2)}, \mathbf{Z}_2^{(i)} = \sigma_z^{(i,2)}$$

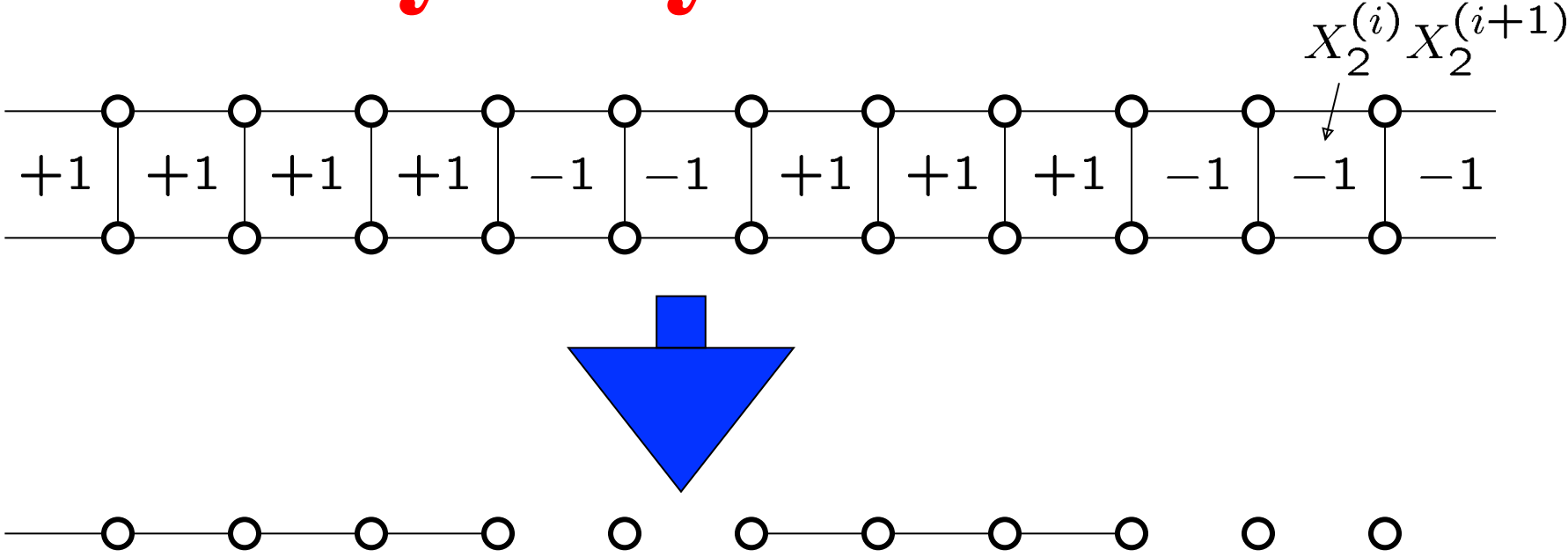
$$\mathbf{H} = -\omega\left(\sum_{i=1}^{n}\mathbf{Z}_1^{(i)} + \sum_{i=1}^{n-1}\mathbf{X}_1^{(i)}\mathbf{X}_1^{(i+1)}\left(\mathbf{I} + \mathbf{X}_2^{(i)}\mathbf{X}_2^{(i+1)}\right)\right)$$

basis $|\vec{Z}_1, \vec{X}_2\rangle = |Z_1^{(1)}, \ldots, Z_1^{(n)}, X_2^{(1)}, \ldots, X_2^{(n)}\rangle$ $\quad Z_1^{(i)}, X_2^{(i)} \in \{\pm 1\}$

$$\mathbf{H} = -\omega\left(\bigoplus_{\vec{X}_2}\left(\sum_{i=1}^{n}\mathbf{Z}_1^{(i)} + \sum_{i=1}^{n-1}c_i(\vec{X}_2)\mathbf{X}_1^{(i)}\mathbf{X}_1^{(i+1)}\right)\otimes|\vec{X}_2\rangle\langle\vec{X}_2|\right)$$

$$c_i(\vec{X}_2) = (1 + X_2^{(i)}X_2^{(i+1)})$$

Everybody's Favorite



Ising + transverse field

transverse field only

Ising + transverse field

transverse field only

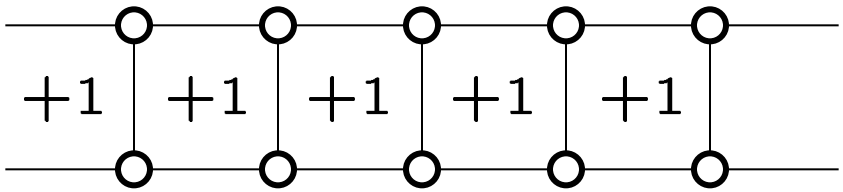
$$\mathbf{H} = -\omega \left(\sum_{i=1}^{n} \mathbf{Z}_1^{(i)} + \sum_{i=1}^{n-1} c_i(\vec{X}_2) \mathbf{X}_1^{(i)} \mathbf{X}_1^{(i+1)} \right) \quad c_i(\vec{X}_2) \in \{0, 2\}$$

Grounded



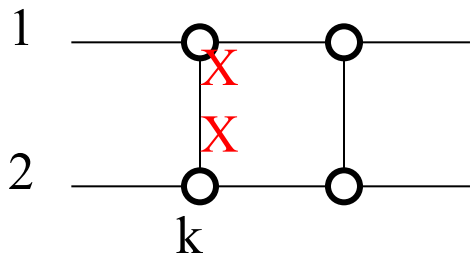
$$c_i(\vec{X}_2) = 2, \forall i$$

Ground state is two-fold degenerate



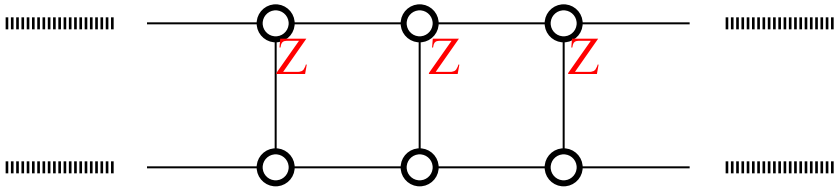
$$\mathbf{X}_2^{(i)} = +1, \forall k$$

$$\mathbf{X}_2^{(i)} = -1, \forall k$$



$$\bar{\mathbf{X}} = \mathbf{X}_2^{(k)}$$

$$[\mathbf{H}, \bar{\mathbf{X}}] = [\mathbf{H}, \bar{\mathbf{Z}}] = 0$$



$$\bar{\mathbf{Z}} = \Pi_{i=1}^{n} \mathbf{Z}_2^{(i)}$$

Detect and Correct

$$(\mathbb{C}^2)^{2n} = \mathcal{H} = \mathcal{H}_s \otimes \mathcal{H}_h \otimes \mathcal{H}_l$$

$$\mathcal{B}(\mathcal{H}) = \mathcal{B}_s \otimes \mathcal{B}_h \otimes \mathcal{B}_l$$

$\mathcal{H}_l = (\mathbb{C}^2)$ logical qubit

$\mathcal{H}_h = (\mathbb{C}^2)^n$ Hamiltonian space

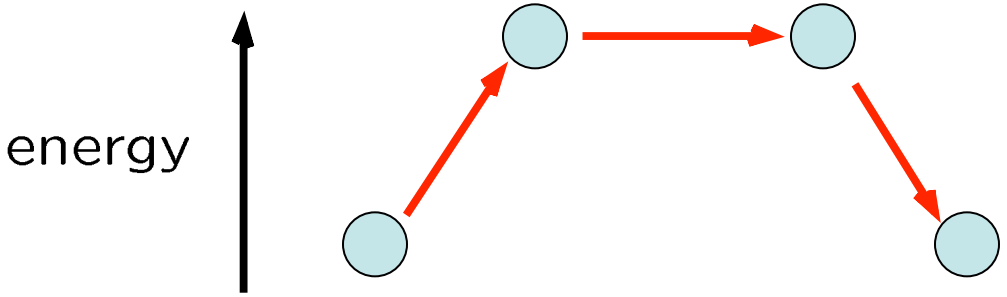
$\mathcal{H}_s = (\mathbb{C}^2)^{n-1}$ stabilizer code

$\bar{\mathbf{X}}, \bar{\mathbf{Z}} \in \mathcal{B}_l \quad \mathbf{X}_2^{(i)}\mathbf{X}_2^{(i+1)} \in \mathcal{B}_s \quad \mathbf{Z}_1^{(i)}, \mathbf{X}_1^{(i)} \in \mathcal{B}_h$

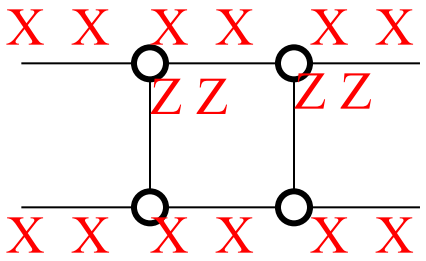
Elements of $\mathcal{B}_s$ imply distance n qubit σ_z detecting code

Can correct $\left\lfloor \frac{n-1}{2} \right\rfloor$, σ_z errors

but only 1 qubit σ_x detecting code

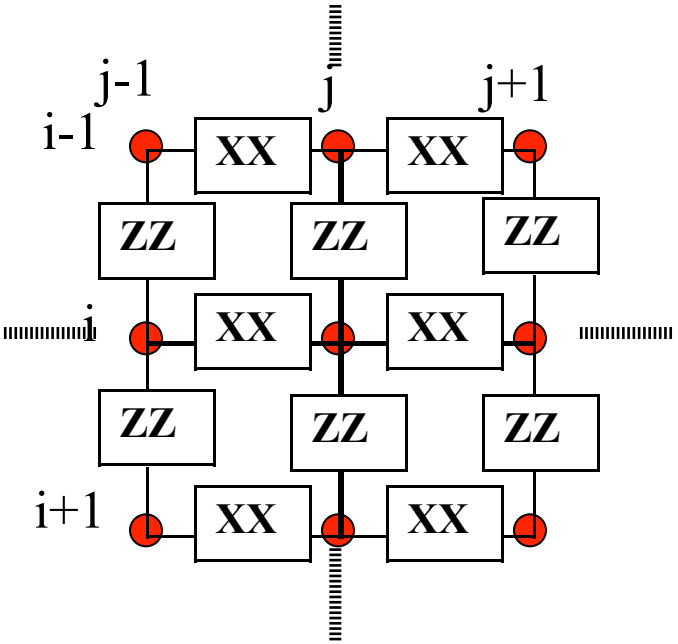


disorders at finite temperature



Into the 2nd Dimension

$$\mathbf{H} = \sum_{i=1}^{n} \sum_{j=1}^{n-1} \sigma_x^{(i,j)} \sigma_x^{(i,j+1)} + \sum_{i=1}^{n-1} \sum_{j=1}^{n} \sigma_z^{(i,j)} \sigma_z^{(i+1,j)}$$



$$[\mathbf{H}, \bar{\mathbf{X}}] = [\mathbf{H}, \bar{\mathbf{Z}}] = 0$$

Look at group algebra generated by individual bounds

Abelian invariant algebra generated double rows:

$$\mathbf{Z}_k = \prod_{j=1}^{n} \sigma_z{}^{(k,j)} \sigma_z^{(k+1,j)} \qquad \mathbf{X}_k = \prod_{i=1}^{n} \sigma_x{}^{(i,k)} \sigma_x^{(i,k+1)}$$

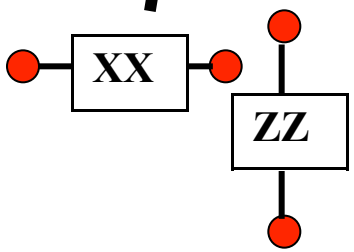
Nonabelian commutant: $\bar{\mathbf{Z}} = \prod_{j=1}^{n} \sigma_z{}^{(k,j)} \quad \bar{\mathbf{X}} = \prod_{i=1}^{n} \sigma_x{}^{(i,k)}$

Tic-Tac-"DOH!"

$$\mathcal{B}(\mathcal{H}) = \mathcal{B}_s \otimes \mathcal{B}_h \otimes \mathcal{B}_l$$

$$\mathbf{Z}_k = \prod_{j=1}^{n} {\sigma_z}^{(k,j)} {\sigma_z}^{(k+1,j)}$$

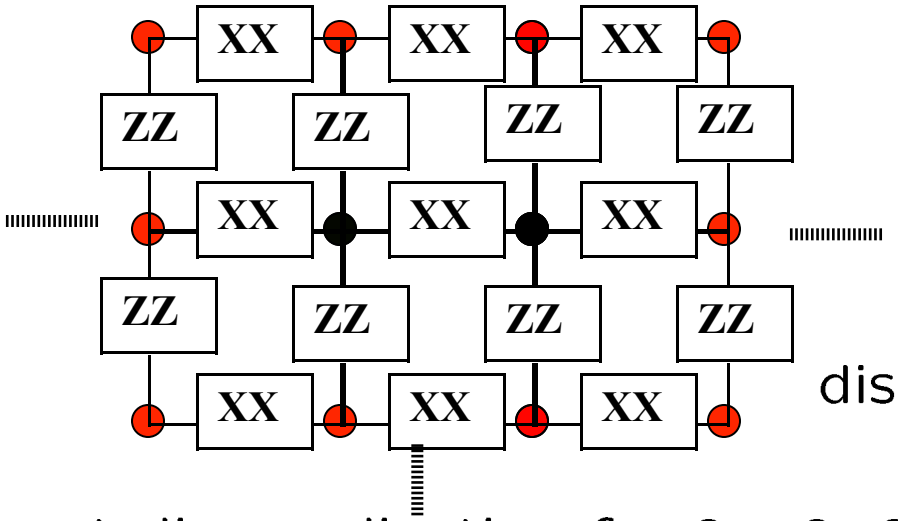
$$\mathbf{X}_k = \prod_{i=1}^{n} {\sigma_x}^{(i,k)} {\sigma_x}^{(i,k+1)}$$

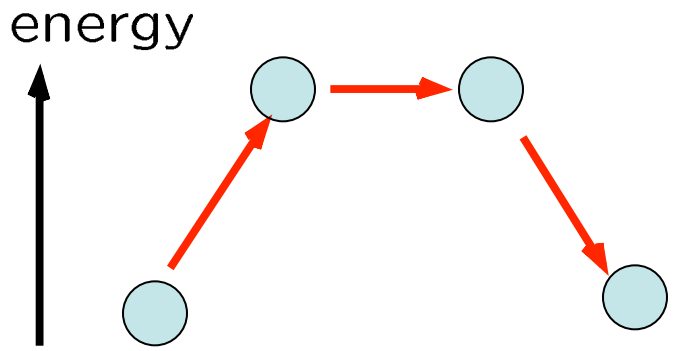


$$\bar{\mathbf{Z}} = \prod_{j=1}^{n} {\sigma_z}^{(l,j)}$$

$$\bar{\mathbf{X}} = \prod_{i=1}^{n} {\sigma_x}^{(i,l)}$$

σ_z errors

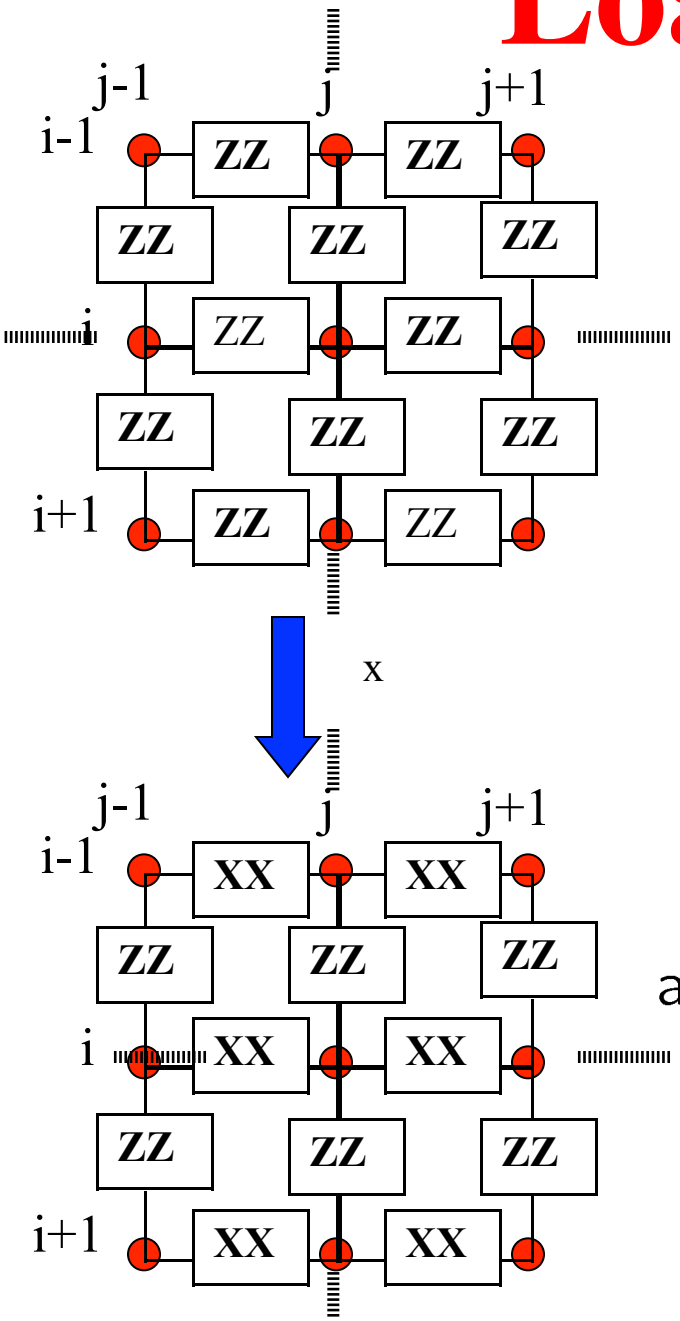


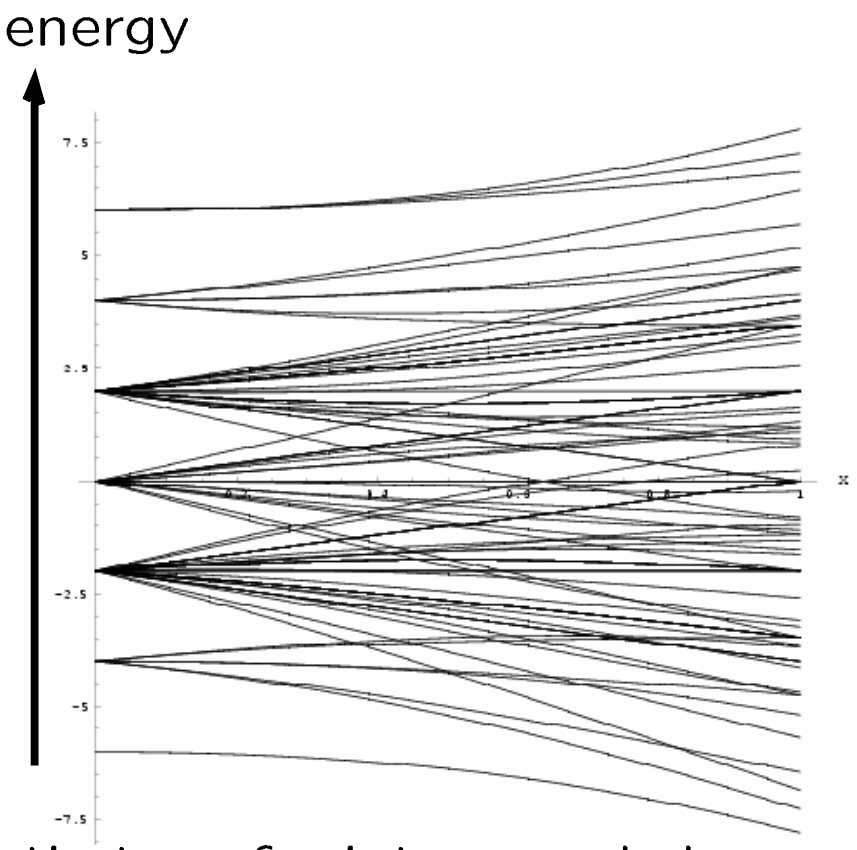


disorders at finite temperature

exact diagonalization for 3×3, 3×4, 4×4, 4×5 supports

Loading 3x3



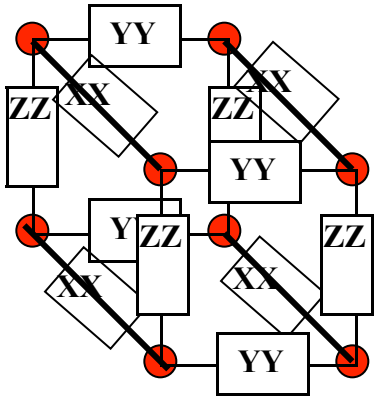


adiabatic transfer into encoded ground state

Into the Third Dimension

$n \times n \times n$

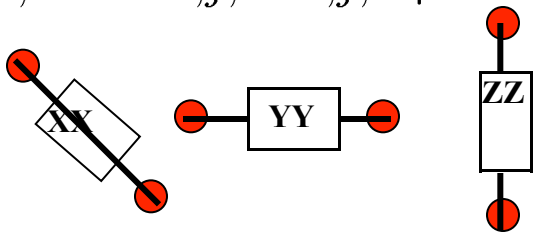
n odd



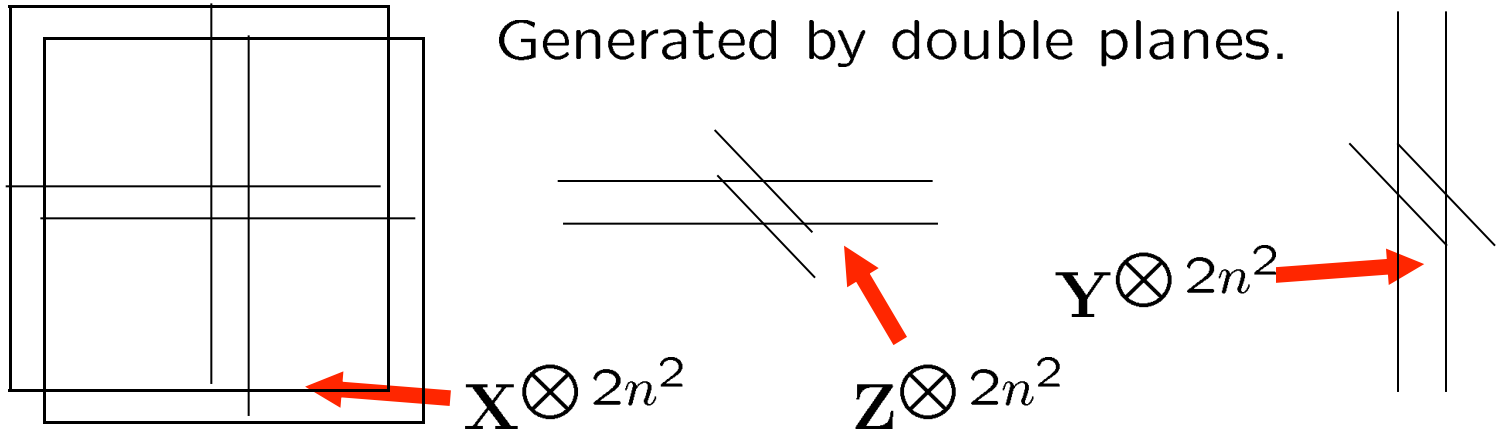
Cubic lattice with qubits at vertices

$$\mathbf{H} = -J \sum_{ijk} (\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{Y}_{i,j,k}\mathbf{Y}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1})$$

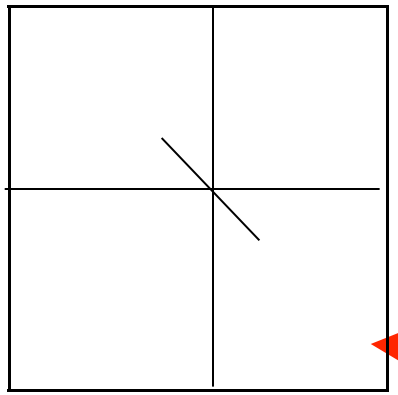
Look at group algebra generated by bonds:



Abelian invariant subalgebra:



Logical Operators



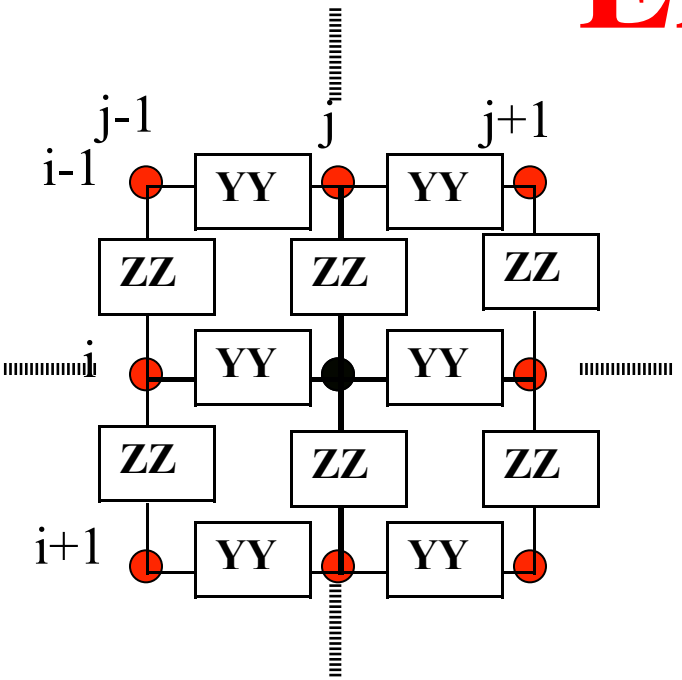
$$\bar{\mathbf{Z}} = \mathbf{Z}^{\otimes n^2}$$

$$\bar{\mathbf{X}} = \mathbf{X}^{\otimes n^2}$$

$$\bar{\mathbf{Y}} = \mathbf{Y}^{\otimes n^2}$$

$$[\bar{\mathbf{X}}, \mathbf{H}] = [\bar{\mathbf{Y}}, \mathbf{H}] = [\bar{\mathbf{Z}}, \mathbf{H}] = 0$$

Errors



σ_z error

σ_y errors stringlike

σ_x errors stringlike

σ_z errors stringlike

Order Parameter and Beyond

$$\mathcal{B}(\mathcal{H}) = \mathcal{B}_s \otimes \mathcal{B}_h \otimes \mathcal{B}_l$$

stabilizer — Hamiltonian — Logical qubit

Subsystem code:

error → diagnose and fix

Error in $\mathcal{B}$ → makes effect idenity on $\mathcal{B}_s, \mathcal{B}_l$

error left on $\mathcal{B}_h$

don't care (tensor product)

This allows us to write down a subsystem order parameter.

Current project: Quantum Monte Carlo for this order parameter.

Fin

Error correcting codes

- classical
- quantum

Fault-tolerance

- local structures giving rise to global robustness (CA)

Representation theory

- identification of qubits and logic



Geometry, Topology

Statistical Physics

- theory of strongly interacting quantum systems

Real Physics

- physical implementation
- optical lattices, squids, donor electrons, etc.

Naturally Fault-Tolerant Quantum Computation